

LMMCoDrive: Cooperative Driving with Large Multimodal Models

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Abstract—To address the intricate challenges of cooperative scheduling and motion planning in Autonomous Mobility-on-Demand (AMoD) systems, this paper introduces LMMCoDrive, a novel cooperative driving framework that leverages a Large Multimodal Model (LMM) to improve traffic efficiency and passenger experience in dynamic urban environments. This framework seamlessly integrates scheduling and motion planning processes to ensure the effective operation of Cooperative Autonomous Vehicles (CAVs). The spatial relationship between CAVs and passenger requests is abstracted into a Bird's-Eye View (BEV) image to fully exploit the potential of the multimodal understanding ability of LMMs. Besides, trajectories are cautiously refined for each CAV while ensuring collision avoidance through safety constraints. A decentralized optimization strategy, facilitated by the Alternating Direction Method of Multipliers (ADMM) within the LMM framework, is proposed to drive the graph evolution of CAVs. Simulation results in diverse urban scenarios demonstrate the pivotal role and significant impact of LMM in optimizing CAV scheduling and seamlessly serving a decentralized cooperative optimization process for each CAV. This marks a substantial stride towards practical, efficient, and safe AMoD systems that are poised to revolutionize urban transportation. The code is available at <https://github.com/henryhcliu/LMMCoDrive>.

I. INTRODUCTION

Autonomous Mobility-on-Demand (AMoD) Systems represent a transformative approach to urban transportation, which aim to efficiently and safely navigate complex urban environments to fulfill passenger requests with fleets of Cooperative Autonomous Vehicles (CAVs) [1]. These systems promise to revolutionize urban landscapes by mitigating traffic congestion, enhancing accessibility, and reducing transportation-related emissions. However, orchestrating such systems presents significant challenges, particularly in the realm of task scheduling and cooperative motion planning [2]. The inherent dynamic nature of AMoD systems necessitates continuous path adjustments for CAVs to

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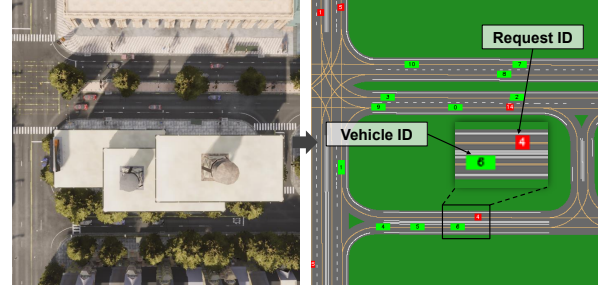


Fig. 1. Demonstration of the autonomous mobility-on-demand system in an urban scenario. The red squares denote the requests from passengers, while the free vehicles are supposed to be scheduled by the LMM. The abstracted bird-eye-view graphics are generated along with the supplementary textual information to be sent to the LMM for multiple decisions.

manage both assigned and future tasks, thereby complicating vehicle scheduling in response to passenger requests within the intricate configurations of urban traffic systems [3].

Historically, research has been carried out to segregate the focus on scheduling, particularly on how CAVs are dispatched to meet user demands. Yet, the critical need for the integration of scheduling and motion planning is neglected. This separation overlooks the essential requirement for simultaneous motion planning and mutual avoidance among assigned vehicles, which is crucial for the practical deployment of AMoD systems in complex urban environments. This research gap signifies a pressing need for possible integrated solutions.

The challenges of integrating intelligent scheduling with real-time cooperative motion planning for AMoD systems are threefold. First and foremost, there is a formidable challenge of developing a universally applicable scheduling strategy that can effectively navigate the intricacies of urban traffic systems, particular for rule-based and learning-based methods [4]. Secondly, ensuring an agile response from the motion planning level to changes at the scheduling level is essential for the dynamic urban traffic environment [5]. Lastly, achieving fast cooperative motion planning with a large number of CAVs, without compromising on safety or efficiency, introduces another level of complexity [6]. In light of these challenges, this work introduces LMMCoDrive, the first approach leveraging a Large Multimodal Model (LMM) to enhance decentralized cooperative scheduling and motion planning of AMoD systems. Our primary contributions are as follows:

- We propose a novel LMM-enhanced framework that integrates scheduling and motion planning processes for CAVs, in which we leverage an informative BEV of

the AMoD system to exploit the potential of LMMs in cooperative driving tasks.

- We propose a decentralized optimization algorithm utilizing the Alternating Direction Method of Multipliers (ADMM) as an integrated motion planner following the LMM-enabled decision-making process. It efficiently divides the CAVs into several sub-graphs, leading to smaller-sized Optimal Control Problems (OCPs) for high computational efficiency.
- Experimental results demonstrate the effectiveness of LMMCoDrive in optimizing the scheduling of CAVs and guiding the motion planning process for the cooperative driving task. The results attained underscore a significant advancement for LMMs towards the realization of practical AMoD systems.

II. RELATED WORKS

A. Scheduling Methods for AMoD

Recent studies have delved into the application of Reinforcement Learning (RL) in the realm of scheduling for AMoD systems. For example, the integration of multi-agent Soft Actor-Critic with weighted bipartite matching achieves small-scale scheduling of the agents [7]. Despite their innovative strategies, RL-based methods grapple with inherent drawbacks. The complexity of AMoD scheduling, which encompasses numerous variables and scenarios, makes it challenging to define a clear and effective reward function [8]. Furthermore, the scarcity of real-world datasets for task scheduling of AMoD systems limits the practical applicability and training of RL models. To navigate the request allocation process, various heuristic methods have been proposed, such as Distance First (DF), Idle First (IF), and Priority First (PF) [9], [10]. In addition, the First-Come First-Serve (FCFS) mechanism is generally used in practice [11], but this policy may decrease traffic efficiency. These methods, guided by predefined rules, offer simplicity and interpretability but may lack the flexibility needed to address the inherent dynamic nature of urban transportation systems, e.g., balancing the waiting time of passengers and the overall traffic efficiency.

On the contrary, The advent of Large Language Model (LLM) has paved the way for advancements in task scheduling, owing to their prowess in open-world comprehension, reasoning, and few-shot acquisition of common sense knowledge [12]–[14]. Notable efforts like DiLu leverage textual descriptions of the environment for single-vehicle decision-making [15]. The potential of LLM to make contextual inferences and decisions heralds a promising direction for scheduling of AMoD systems. Nonetheless, the application of multimodal LLM or LMM in the scheduling or cooperative motion planning for AMoD systems has yet to be extensively explored [16]. With their robust information representation capabilities, especially in dealing with images, LMMs have the potential to greatly improve the depiction and comprehension of urban transportation systems, which could underscore a crucial research gap.

B. Cooperative Motion Planning Methods for AMoD

After the scheduling process of the AMoD system, each CAV is required to plan a feasible trajectory to be followed. Optimization-based approaches to cooperative motion planning typically employ a formulation as an OCP. This methodological framework offers a precise mathematical representation, strong interpretability, and guarantee of optimality [17], [18]. For CAVs with nonlinear vehicle models, OCPs are typically solved using established nonlinear programming solvers, such as interior point optimizer (IPOPT) and Sequential Quadratic Programming (SQP). Moreover, the iterative linear quadratic regulator (iLQR) method, benefiting from Differential Dynamic Programming (DDP), addresses nonlinear optimization problems by retaining only the first-order term of dynamics through Gauss-Newton approximation [19]. However, traditional iLQR struggles to directly manage inequality constraints related to collision avoidance and physical limitations in this situation. To address this challenge, advanced variations such as control-limited DDP and constrained iLQR have been developed [20]–[22].

To alleviate the computational burdens of cooperative driving in large-scale AMoD systems, ADMM emerges as a potent solution by decomposing the principal optimization problem into several sub-problems [23]. Its parallel and distributed nature renders ADMM particularly suitable for cooperative motion planning of CAVs. For instance, dual consensus ADMM has been deployed to facilitate a fully parallel optimization framework for cooperative motion planning of CAVs [24]. This approach significantly distributes computational efforts across all entities and attains real-time performance. Despite these advancements, existing methods fall short of supporting large-scale cooperative driving due to the fully connected nature of agents. Our prior research, which leverages the sparsity attribute of optimization problems for involved vehicles [25], indicates a promising pathway to enhance computational efficiency for AMoD systems, and this highlights another research gap to be addressed.

III. PROBLEM FORMULATION

We consider a discrete-time system with N single-occupancy vehicles, denoted by $\mathcal{V}$, and M potential passengers described as $\mathcal{M}$. Passengers arrive at random locations and time steps along the road network and wait for transportation to their respective destinations. Unlike traditional heuristic methods, our designed cooperative driving scheme for AMoD systems determines the service order, comprehensively prioritizing transport efficiency, passenger experience, energy saving, and other factors considered in the AMoD systems. The representation of CAVs' relationships using graph theory, where nodes and edges signify vehicles and communication links respectively, facilitates a detailed analysis of their interactions. We define an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, with $\mathcal{E}$ as the communication links. Each vehicle $n^i \in \mathcal{V}$ is characterized by a state vector $\mathbf{z}^i$ and a control input vector $\mathbf{u}^i$, both derived from the vehicle kinematics. Subgraphs $\mathcal{H}$, which represent the graphs of a set of vehicles with collision risks, evolve over planning horizons. An edge

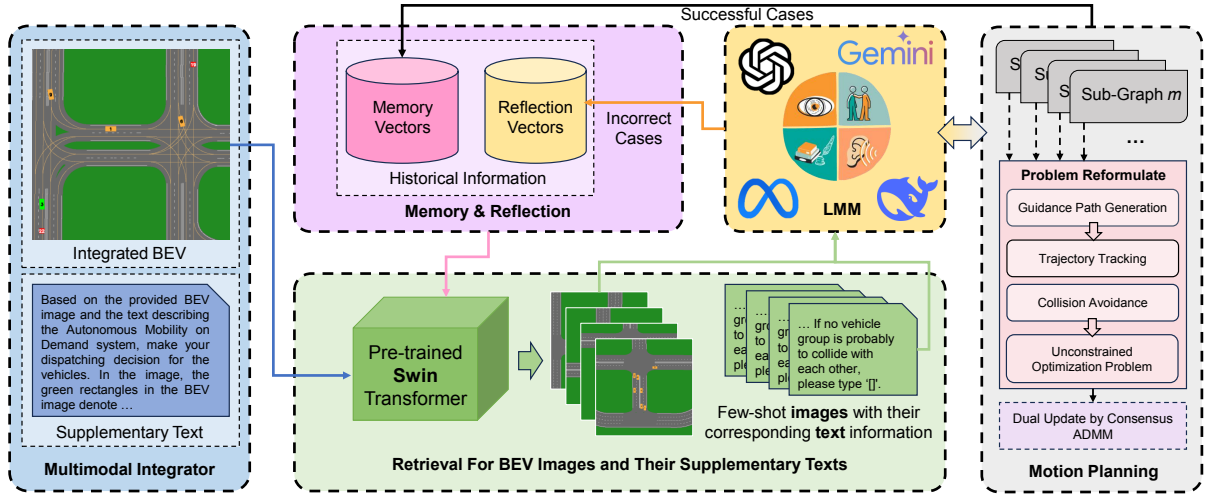


Fig. 2. Overall architecture of LMMCoDrive. It is composed of a multimodal integrator, a memory and reflection module, a retrieval module for similar memory and reflections, and a decentralized cooperative driving module. CMP means cooperative motion planning for the CAVs in the AMoD system.

$(n^i, n^j) = d^{i,j} \in \mathcal{E}_h \subseteq \mathcal{E}$ exists if the inter-vehicle distance $d^{i,j}$ is less than the communication range r_{tele}^i . The degree of a node n^i , i.e., $|n^i| = \deg(n^i)$, indicates its connectivity with the nodes in the set

$$\mathcal{N}^i = \{j \in \mathcal{N} \mid (i, j) \in \mathcal{E}_h\}. \quad (1)$$

With the preceding analysis and definitions, it aims to efficiently and economically deliver all passengers in $\mathcal{M}$ using a limited number of CAVs in $\mathcal{V}$, while ensuring acceptable waiting time for passengers. The vehicles within $\mathcal{H}$ must cooperatively plan their trajectories to avoid collisions. Consequently, the AMoD system operates seamlessly in terms of task allocation and passenger delivery.

IV. METHOD

Our proposed LMMCoDrive scheme mainly contains two parts: First, an LMM-based scheduling agent, is responsible for the multimodal information gathering, memory, reflection, and retrieval of similar historical situations. Second, an LMM-guided cooperative driving module, provides low-level motion planning for the vehicles and ensures safety. The executing process is exhibited in Fig. 2.

A. LMM-Based Scheduling

The proposed LMM-based scheduling framework is designed to integrate and process multimodal information for effective task scheduling of AMoD systems. This section delves into the framework's core modules: Multimodal Information Integrator, Memory and Reflection Module, and Retrieval Module.

1) *Multimodal Integrator*: The multimodal integrator is pivotal in synthesizing data from various sources to generate a comprehensive understanding of the urban traffic environment and AMoD system status. This module comprises two primary components: BEV image generation and supplementary text generation.



Fig. 3. The supplementary text (system message and human message) for the BEV image to enhance the understanding ability of LMMs. Different colors of the text demonstrate different purposes of each text chunk.

This module starts by creating BEV images that encapsulate critical elements of the urban traffic landscape. These elements include road boundaries, lanes, CAVs each tagged with a unique ID, the centerlines of road lanes, destinations associated with each CAV (identified by their respective IDs), and pending requests from passengers marked in alphabetical order. The creation of BEV images is instrumental in providing a visual-grounded foundation for the scheduling

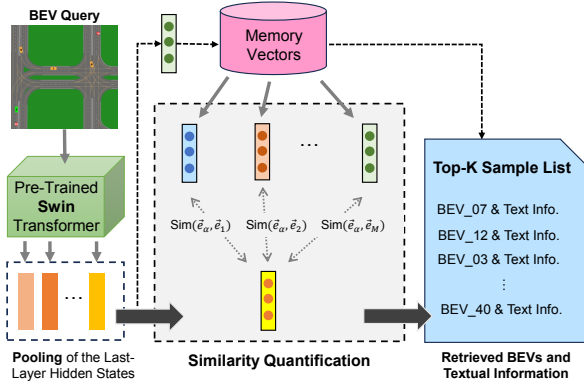


Fig. 4. Illustration of the retrieval for LMMCoDrive. A BEV image for similarity query is embedded with the pre-trained swin-transformer, before the pair-wise similarity quantification with the memory vectors. The memory samples of Top-K similarities are retrieved to support the online reasoning.

system's decision-making processes.

Following the generation of BEV images, the module undertakes the production of supplementary text as shown in Fig. 3. This text plays multiple roles, serving first as a *SystemMessage* that offers a deeper dive into the driving environment. It differentiates types of lane markings and delineates the driving layout. Specifically, as demonstrated in the *integrated BEV* of Fig. 2, the CAVs and unresponded passenger requests are identified by exclusive IDs, enabling the grounding accuracy of textual reasoning.

2) *Memory and Retrieval for Enhanced CAV Cooperation*: As shown in the middle of Fig. 2, this module consists of two parts: the vectors of historical information, and the retrieval utility for similar historical multimodal information. The historical information contains successful memory vectors and reflected incorrect vectors act as a repository, storing historical images along with their corresponding textual information (i.e. prompting text and the response from the LMM) in a stack. The i th item of the storage can be expressed as

$$M_i = \{\mathcal{I}_i, \mathcal{H}_i, \mathcal{R}_i, \vec{e}_i\}, \quad (2)$$

where $\mathcal{I}_i$ is the BEV image, $\mathcal{H}_i$ is the supplementary text for $\mathcal{I}_i$, $\mathcal{R}_i$ is the corresponding response from the LMM. For fast retrieval, the feature embedding $\vec{e}_i = \mathcal{K}(\mathcal{I}_i)$ is also stored in the storage item. Note that $\mathcal{K}(\cdot)$ is the neural network to generate the image embeddings. This stack is limited in size and stored in vectorized format, ensuring that only the most recent and relevant information is swiftly requested for future reference.

Moreover, to facilitate few-shot inference for scheduling decisions, the storage items M_i similar to the current context should be retrieved with an appropriate approach. In the framework of LMMCoDrive, each BEV image in the memory storage is assigned an embedding using a pre-trained vision model. In this study, we adopt the Swin-Transformer [26], pre-trained on ImageNet-21k [27], owing to its superior capability in modeling multi-scale visual-grounded entities. This allows for the effective extraction

of features pertinent to both macroscopic traffic scenarios (such as intersections and multi-lane roadways) and microscopic object statuses (including vehicle IDs, positions, and orientations). The embedding is derived by average pooling of the final-layer hidden states to summarize the global feature representation. As illustrated in Fig. 4, during online retrieval, the real-time BEV image is first embedded using the vision model, and then its similarity to the memory samples is evaluated using the cosine similarity score:

$$\text{Sim}(\vec{e}_\alpha, \vec{e}_\beta) = \frac{\vec{e}_\alpha \cdot \vec{e}_\beta}{\|\vec{e}_\alpha\| \|\vec{e}_\beta\|}, \forall \beta \in \{1, 2, \dots, M\}, \quad (3)$$

where $\vec{e}_\alpha$ denotes the embedding of the newly generated BEV image used for the similarity query, while $\vec{e}_\beta$ represents the embedding of any BEV image stored in the Memory module, and M indicates the size of the memory. Following this, the BEV images with the top- K similarities are retrieved. This multimodal information forms the few-shot data that guides the online reasoning process.

3) *Multimodal Reasoning for Cooperative Scheduling*: When the adaptive scheduling mechanism activates, our LMM agent generates cooperative dispatching assignments through structured reasoning. The system employs chain-of-thought (CoT) [12] processing to ensure interpretable decision-making, formulated as:

$$\mathcal{A}_\tau, \mathcal{R}_\tau = \Psi(\Gamma(\mathcal{I}_\tau, \Omega_\tau, \mathcal{H}_\tau)) \quad (4)$$

where $\Gamma(\cdot)$ integrates three knowledge sources: foundational instructions $\mathcal{I}_\tau$, dynamic memory states Ω_τ , and multimodal observations $\mathcal{D}_\tau$. The reasoning operator $\Psi(\cdot)$ produces dual outputs per time step τ : vehicle-request assignment pairs $\mathcal{A}_\tau = \{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\}$, $m = \max(N_{\text{free}, \tau}, M_\tau)$ and structured reasoning traces $\mathcal{R}_\tau$. Note that $N_{\text{free}, \tau}$ and M_τ denote the numbers of current free vehicles x_i and the un-responded passengers y_i , respectively. This dual-output architecture enhances operational transparency while providing essential guidance for scheduling. Specifically, the derived assignments $\mathcal{A}_\tau$ subsequently guide multi-agent trajectory coordination through our hierarchical cooperative driving framework, ensuring coherent execution of passenger service tasks across CAVs. This tight integration between high-level cooperation logic and motion planning (see Section IV-B) constitutes an efficient and low-cost cooperation framework for AMoD systems.

B. LMM-Guided Cooperative Motion Planning

As illustrated in Fig. 2, the motion planning problem is reformulated after receiving the task allocation decision for the free vehicles from LMMs. Specifically, this decision is passed to the motion planning module, which then generates a global rough path for the assigned vehicles using search-based methods supported by the proposed platform. Following the problem reformulation, solving this cooperative motion planning problem becomes challenging due to the large-scale optimization variables involved. In this paper, we leverage dual consensus ADMM for the OCP decomposition and parallel optimization.

Algorithm 1 Decentralized Optimization via Consensus ADMM in One Subgraph $\mathcal{H}(\mathcal{N}, \mathcal{E}_h)$

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1: initialize  $\{z_\tau^i, u_\tau^i\}_{\tau=0}^T, \{p^{i,0}, y^{i,0}, z^{i,0}, s^{i,0}\}, \forall i \in \mathcal{N}$ 
2: choose  $\sigma, \rho > 0$ 
3: repeat:
4:   Send  $\{z_\tau^i\}_{\tau=1}^T$ , receive  $\{z_\tau^j\}_{\tau=1}^T$  from  $j \in \mathcal{N}^i$ 
5:   Compute  $k^i, J^i, \{A_\tau^i\}_{\tau=0}^{T-1}, \{B_\tau^i\}_{\tau=0}^{T-1}$ 
6:   reset  $p^{i,0} = s^{i,0} = 0, y^{i,0} = y^{\text{last}}, x^{i,0} = x^{\text{last}}$ 
7:   reset  $y_{[2]}^{i,0} = x_{[2]}^{i,0} = 0$ 
8:   repeat: for all  $i \in \mathcal{N}$ 
9:     Send  $y^{i,k}$ , receive  $y^{j,k}$  from  $j \in \mathcal{N}^i$ 
10:    Broadcast  $y^{i,k}$  to the vehicles in  $\mathcal{N}^i$ 
11:     $p^{i,k+1} = p^{i,k} + \rho \sum_{j \in \mathcal{N}^i} (y^{i,k} - y^{j,k})$ 
12:     $s^{i,k+1} = s^{i,k} + \sigma (y^{i,k} - x^{i,k})$ 
13:     $r^{i,k+1} = \sigma x^{i,k} + \rho \sum_{j \in \mathcal{N}^i} (y^{i,k} + y^{j,k}) - (k^i + p^{i,k+1} + s^{i,k+1})$ 
14:    Perform (8a)-(8c) for  $\alpha_{[2,i]}^{i,k}, \alpha \in \{p, s, r\}$ 
15:    Perform (9a)-(9c) for  $\alpha_{[2,j]}^{i,k}, \alpha \in \{p, s, r\}, j \in \mathcal{N}^i$ 
16:    Compute  $z^{i,k+1}$  by solving LQR problem (10)
17:     $y_{[1]}^{i,k+1} = 2\gamma \left( \hat{J}^i z_{[1]}^{i,k+1} + r_{[1]}^{i,k+1} \right)$ 
18:     $x_{[1]}^{i,k+1} = \Pi_{\mathcal{R}^+} \left( \frac{1}{\sigma} s_{[1]}^{i,k+1} + y_{[1]}^{i,k+1} \right)$ 
19:    Perform (8d-8e) for  $\alpha_{[2,i]}^{i,k}, \alpha \in \{y, x\}$ 
20:    Perform (9d-9e) for  $\alpha_{[2,j]}^{i,k}, \alpha \in \{y, x\}, j \in \mathcal{N}^i$ 
21:     $k = k + 1$ 
22:  until number of iteration steps exceeds  $k_{\max}$ 
23:  Update  $\{z_\tau^i, u_\tau^i\}_{\tau=0}^T$ 
24: until termination criterion is satisfied

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For the allocated AMoD vehicle fleet, we consider a cooperative motion planning task within a subgraph $\mathcal{H} \in \mathcal{G}$ generated by conditional Manhattan distance. We formulate this task, involving $N_{\mathcal{H}}$ CAVs on urban roads with specified destinations and paths, as an OCP:

$$\begin{aligned}
& \min_{z_\tau^i, u_\tau^i} \sum_{i=1}^{N_{\mathcal{H}}} Q_i(Z^i, U^i) \\
& \text{s.t.} \quad z_{\tau+1}^i = f(z_\tau^i, u_\tau^i), \\
& \quad z_{\tau+1}^i \in \mathcal{S}_\tau^i, \\
& \quad -\underline{u}_\tau^i \preceq u_\tau^i \preceq \bar{u}_\tau^i, \\
& \quad \forall \tau \in \mathcal{T}, \forall i \in \mathcal{V}_{\mathcal{H}},
\end{aligned} \tag{5}$$

where the concatenated state and input variables are

$$Z^i = [z_0^i, z_1^i, \dots, z_T^i], U^i = [u_0^i, u_1^i, \dots, u_{T-1}^i].$$

Note that $\mathcal{V}_{\mathcal{H}}$ represents the set of indices of CAVs within a subgraph $\mathcal{H}$, and $\mathcal{T} = \{0, 1, \dots, T-1\}$ denotes the temporal horizon for one cooperative motion planning episode. $\mathcal{S}_\tau^i$ defines the collision-free and bounded operational region for the i th CAV at time step τ . $\underline{u}^i$ and $\bar{u}^i$ represent the lower and upper bounds of the control input for the i th CAV, respectively. The objective of this OCP is to enable the CAVs to follow their reference paths assigned by the upstream LMM-enabled scheduling decisions. For an individual vehicle, this

Algorithm 2 Cooperative Scheduling and Motion Planning for CAVs with LMM

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1: schedule target positions of all CAVs in  $\mathcal{G}$  using LMM
2: determine optimal route using A* for each CAV
3: smooth the waypoints in each route by the Savitzky-Golay filter
4: while  $\exists$  requests in the AMoD do:
5:   Execute graph evolution to distribute the CAVs to the subgraphs  $\mathcal{H}^k, k \in \mathbb{R}_+$ 
6:   for subgraph  $\mathcal{H}^i$  in subgraphs list do
7:     search nearest waypoints for each CAV at each time step  $\tau \in \{0, 1, \dots, T_s\}$  using KD-tree
8:     Execute Algorithm 1 to obtain CAVs' trajectories
9:   end for
10:  perform first  $T_e$  steps of the CAVs in  $\mathcal{G}$ 
11:  feedback at time step  $T_e$ 
12:  relay the states of all the CAVs in  $\mathcal{G}$ 
13: end while

```

can be expressed as:

$$Q_i(z_\tau^i, u_\tau^i) = \sum_{\tau=0}^T \|z_\tau^i - z_{\text{ref},\tau}^i\|_Q^2 + \sum_{\tau=0}^{T-1} \|u_\tau^i\|_R^2, \tag{6}$$

where Q and R are the weighting matrices balancing the path-tracking and energy-saving indicators.

In order to reduce the size of the OCP and provide an agile response, Model Predictive Control (MPC) is used with a planning horizon of $T_p = 15$ and an executing horizon of $T_e = 10$. To formulate a tractable problem, using second-order Taylor expansion and local linearization, the dual formulation of the OCP derived from (5) is

$$\begin{aligned}
& \min_{\Delta Z^1, \dots, \Delta Z^N} \sum_{i=1}^N F^i(\Delta Z^i) + \mathcal{I}_{\mathcal{K}}(h) \\
& \text{s.t.} \quad \sum_{i=1}^N (J^i \Delta Z^i - k^i) = h,
\end{aligned} \tag{7}$$

where ΔZ^i is the perturbed vector of the above state and control variables sequentially. F^i is the host cost for vehicle i , $\mathcal{I}_{\mathcal{K}}$ is the indicator function for the mutual avoidance and bounded constraints. Leveraging the Lagrangian of (7), the cooperative motion planning of the vehicles in $\mathcal{H}$ can be derived with the dual variables p, s, r, y, x . As shown in Algorithm 1, *dual update* for the dual variables is achieved by lines 10-15 and 17-20, while *primal update* is realized using line 16. Due to space constraints, a similar derivation process can be found in [28].

To enhance computational efficiency in the dual update process, we define two distinct sets of variables: one set associated with the target vehicle and another set pertaining to the surrounding vehicles that pose a collision risk to the target vehicle. We leverage (8) and (9) for the target vehicle and its surrounding connected vehicles, respectively. Note that for the dual update of $y_{[2,i]}^{v,k+1}$ of the surrounding vehicles connected to the target vehicle, they only need to consider

the consensus to $\mathbf{r}_{[2,i]}^{k+1}$ in (9d). For comparison, the target vehicle needs to consider the box constraints by calculating the high-dimensional matrix multiplication in (8d).

$$\mathbf{p}_{[2,i]}^{i,k+1} = \mathbf{p}_{[2,i]}^{i,k} + \rho(N-1) \left(\mathbf{y}_{[2,i]}^{i,k} - \mathbf{y}_{[2,i]}^k \right), \quad (8a)$$

$$\mathbf{s}_{[2,i]}^{i,k+1} = \mathbf{s}_{[2,i]}^{i,k} + \sigma \left(\mathbf{y}_{[2,i]}^{i,k} - \mathbf{x}_{[2,i]}^k \right), \quad (8b)$$

$$\begin{aligned} \mathbf{r}_{[2,i]}^{i,k+1} &= \sigma \mathbf{x}_{[2,i]}^{i,k} + \rho(N-1) \left(\mathbf{y}_{[2,i]}^{i,k} + \mathbf{y}_{[2,i]}^k \right) \\ &\quad - (\mathbf{k}_{[2,i]}^{i,k+1} + \mathbf{p}_{[2,i]}^{i,k+1} + \mathbf{s}_{[2,i]}^{i,k+1}), \end{aligned} \quad (8c)$$

$$\mathbf{y}_{[2,i]}^{i,k+1} = 2\gamma \left(\mathbf{O}^i \Delta \mathbf{Z}^{i,k+1} + \mathbf{r}_{[2,i]}^{i,k+1} \right), \quad (8d)$$

$$\mathbf{x}_{[2,i]}^{i,k+1} = \Pi_{S^{bo}} \left(\frac{1}{\sigma} \left(\mathbf{s}_{[2,i]}^{i,k+1} + \mathbf{y}_{[2,i]}^{i,k+1} \right) \right). \quad (8e)$$

$$\mathbf{p}_{[2,i]}^{v,k+1} = \mathbf{p}_{[2,i]}^k + \rho \left(\mathbf{y}_{[2,i]}^k - \mathbf{y}_{[2,i]}^{i,k} \right) = \mathbf{p}_{[2,i]}^{k+1}, \quad (9a)$$

$$\mathbf{s}_{[2,i]}^{v,k+1} = \mathbf{s}_{[2,i]}^k + \sigma \left(\mathbf{y}_{[2,i]}^k - \mathbf{x}_{[2,i]}^k \right) = \mathbf{s}_{[2,i]}^{k+1}, \quad (9b)$$

$$\begin{aligned} \mathbf{r}_{[2,i]}^{v,k+1} &= \sigma \mathbf{x}_{[2,i]}^k + \rho(2|n^v| - 1) \mathbf{y}_{[2,i]}^k + \rho \mathbf{y}_{[2,i]}^{i,k} \\ &\quad - (\mathbf{k}_{[2,i]}^{k+1} + \mathbf{p}_{[2,i]}^{k+1} + \mathbf{s}_{[2,i]}^{k+1}) = \mathbf{r}_{[2,i]}^{k+1}, \end{aligned} \quad (9c)$$

$$\mathbf{y}_{[2,i]}^{v,k+1} = 2\gamma \mathbf{r}_{[2,i]}^{k+1} = \mathbf{y}_{[2,i]}^{k+1}, \quad (9d)$$

$$\mathbf{x}_{[2,i]}^{v,k+1} = \Pi_{S^{bo}} \left(\frac{1}{\sigma} \left(\mathbf{s}_{[2,i]}^{k+1} + \mathbf{y}_{[2,i]}^{k+1} \right) \right) = \mathbf{x}_{[2,i]}^{k+1}, \quad (9e)$$

For the primal update process, due to the fully decentralized feature of the algorithm, only one target vehicle is considered in a standard LQR problem formulated as:

$$\begin{aligned} \min_{\Delta \mathbf{Z}^i} & \Delta \mathbf{Z}^{i\top} \mathbf{L}_1^i + \frac{1}{2} \Delta \mathbf{Z}^{i\top} \mathbf{L}_2^i \Delta \mathbf{Z}^i + 2\mathbf{r}^{i,k+1\top} \mathbf{J}^i \Delta \mathbf{Z}^i \\ & + \frac{1}{\sigma + 2\rho|n^i|} \Delta \mathbf{Z}^{i\top} \mathbf{J}^{i\top} \mathbf{J}^i \Delta \mathbf{Z}^i \\ \text{s.t.} & (\mathbf{L}_3^i - \mathbf{L}_4^i) \Delta \mathbf{Z}^i = 0. \end{aligned} \quad (10)$$

Furthermore, as depicted in the Algorithm 2, in the scope of task execution, for the vehicles in the AMoD system, the vehicles need to evolve the connection graph and solve the above cooperative motion planning problems guided by the LMM parallelly using a receding horizon manner.

V. SIMULATION RESULTS

A. Environment and Evaluation Metrics

We comprehensively evaluate the proposed LMMCoDrive framework in an AMoD system. For the scheduling part of the pairs of passengers and vehicles, we compare our proposed method with several popular heuristic algorithms, First Come First Serve (FCFS) and Distance First (DF). As shown in Fig. 5, all the experiments are executed in the two urban configurations of CARLA [29] with the assistance of LangChain in Python. The evaluation metrics are divided into two categories: scheduling and cooperative driving. For scheduling performance, we utilize the task completion time T_{tc} to assess the overall effectiveness of LMMCoDrive. Specifically, we calculate the average task completion time T_{atc} , the standard deviation T_{stc} , and the maximum task completion time T_{mtc} . The computational time for each



(a) Top-down view of Town10 (b) Top-down view of Town02

Fig. 5. The illustration of the two representative urban cooperative driving scenarios, Town10 and Town02 in CARLA. (a) The road is wide and many intersections exist. (b) The map contains many one-way road and less connections compared with Town10.

TABLE I
PARAMETER SETTINGS OF THE SIMULATION

param.	value	param.	value	param.	value
Town	02, 10	σ	0.05	$\mathcal{K}(\cdot)$	Swin
p_{img}	1900×1800	ρ	0.002	LMM	GPT-4o
n_{fs}	3	a_{max}	3 m/s ²	C_{Town02}	(100, 210)
scale	8 ppm	a_{min}	-5 m/s ²	C_{Town10}	(-2, 35)

episode of cooperative driving is utilized to represent the effectiveness of the LMM-guided graph evolution strategy. The pertinent parameter settings in the simulation are detailed in Table I. Here, p_{img} represents the resolution of a BEV image, n_{fs} denotes the few-shot number, and the scale of 8 ppm indicates that a length of 8 pixels corresponds to 1 meter in the image. Additionally, Swin is employed to extract feature embeddings from $\mathcal{I}_i$, while GPT-4o serves as the foundational model for LMMCoDrive. For different map configurations, the BEV centers are denoted as $C_{\text{Town } i} \in \mathbb{R}^2$. The remaining parameters are utilized to address the optimization problem outlined in (5).

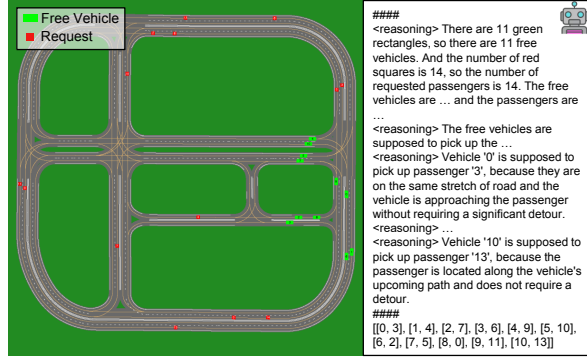
B. Scheduling Performance

We conduct the comparisons in the aforementioned AMoD system with 10 CAVs, and requests from passengers from all over the traffic system are gradually raised. As shown in Fig. 6, the vehicles convey passengers to their randomly generated destinations with the yellow remark. As for the free vehicles, once they are assigned a passenger in the AMoD system, they will drive to their customs and serve them as soon as possible. As depicted in Table II, in most cases, LMMCoDrive has the smallest T_{mtc} and T_{stc} , while maintaining good T_{atc} . It proves the excellent scheduling performance for the AMoD. The LMMCoDrive achieves a balance between efficiency and passenger experience, but performs unstable in some cases, and this indicates that the prompt of the LMM can be further refined.

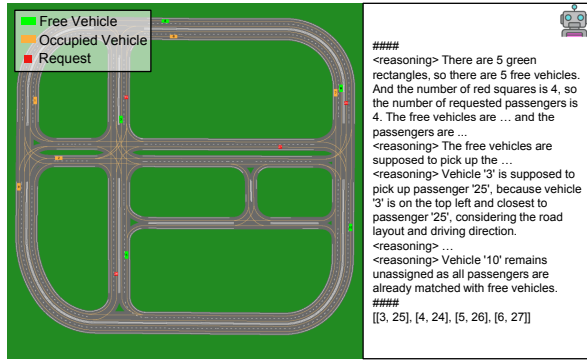
To comprehensively evaluate the robustness of the LMMCoDrive framework, we conducted additional experimental trials across various maps over a time period of 50s, as shown in Table III. The overall traffic efficiency, measured by the number of completed passengers N_{comp} , demonstrates

TABLE II
COMPARISON OF SCHEDULING EFFICIENCY BETWEEN LMMCoDrive AND TWO HEURISTIC ALGORITHMS

Method	Episode 1			Episode 2			Episode 3			Episode 4		
	T_{atc}	T_{stc}	T_{mtc}	T_{atc}	T_{stc}	T_{mtc}	T_{atc}	T_{stc}	T_{mtc}	T_{atc}	T_{stc}	T_{mtc}
FCFS	16.06	5.73	22.78	11.15	5.43	14.64	17.74	3.15	22.98	9.93	8.30	18.24
DF	10.72	6.69	22.97	4.81	5.66	14.61	5.83	4.05	12.31	9.93	8.30	18.24
LMMCoDrive	13.23	5.72	19.78	3.49	3.95	10.34	13.46	3.17	18.23	8.42	7.06	15.48



(a) The Initialized BEV of the AMoD



(b) The BEV of the AMoD after 20 steps

Fig. 6. The reasoning demonstration of the AMoD system utilizing our proposed LMMCoDrive framework. The corresponding responses from the LMM are also revealed alongside the corresponding BEV images.

that the AMoD system operated by the proposed LMMCoDrive outperforms the baseline methods. It is important to note that the 0-shot version of LMMCoDrive shows compromised performance and does not significantly outperform the FCFS and DF methods. Furthermore, the inference time T_{inf} does not significantly increase when memory items are added to the prompt of the LMM.

C. Computational Efficiency after Coordination

The purpose of this experiment is to evaluate the effectiveness of the LMM-generated dispatching decision in cooperative motion planning for AMoD systems. As traffic coordination affects the distribution of CAVs on the map, the constructed OCPs are different and exhibit varying computational burdens. As illustrated in Fig. 7, the LMM's capability to appropriately determine scheduling orders of CAVs results in a significantly lower average computation time for

TABLE III
ROBUSTNESS EVALUATION CROSS DIFFERENCE MAPS

Method	Index	Map	
		Town02	Town10
FCFS	T_{tc} (s)	30.4±15.25	30.85±8.34
	N_{comp}	5	7
DF	T_{tc} (s)	17.0±12.37	20.0±13.16
	N_{comp}	4	6
LMMCoDrive (0-shot)	T_{tc} (s)	32.5±10.14	45.2±2.99
	N_{comp}	4	5
	T_{ref} (s)	8.34±2.91	8.61±3.02
LMMCoDrive	T_{tc} (s)	36.28±10.44	34.00±8.00
	N_{comp}	7	9
	T_{ref} (s)	9.29±1.27	9.40±2.54

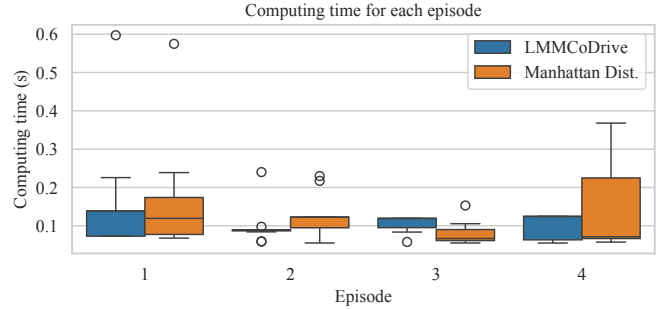


Fig. 7. The computation time for each episode of the receding horizon for cooperative motion planning. For the first column of boxes, the graph of the connection of the vehicles is evolved by LMM, while the second column is the result of the Manhattan distance-based heuristic method.

parallel OCPs compared to the distance-based method. The primary reason for this improvement is that LMMCoDrive can more precisely identify potential congestion, thereby avoiding crowded traffic conditions that would otherwise increase the number of CAVs in one OCP leading to computational overhead. Nonetheless, the prolonged reasoning time of LMMs remains a challenge that must be addressed for the real-time implementation of LMMCoDrive.

D. Hazard Analysis

The generation of scheduling decisions from the LMM may result in inefficiencies and failures for several reasons. First, as illustrated in Fig. 3, the prompt indicates that all available CAVs should be assigned a passenger when $N_{free, \tau} \geq M_{\tau}$. However, there are instances where the LMM outputs fewer CAV-passenger pairs than expected, leading to the neglect of some CAVs and resulting in prolonged idle

parking for several vehicles. Second, the AMoD system is susceptible to crashes due to the receipt of incorrectly formatted decision-making instructions. This incorrect format often arises from unnecessary analyses conducted after the final dispatch decision, which can lead to decoding failures that are then transmitted downstream to the cooperative motion planning process.

VI. CONCLUSION

In this paper, we propose LMMCoDrive, an LMM-enabled integrated scheduling and cooperative motion planning framework suitable for AMoD systems in urban scenarios. The incorporation of a BEV that describes the traffic situation expands the boundary of the applications of LLMs, and this alleviates the travel prompt engineering. Simulation results demonstrate LMM's role in optimizing CAV scheduling and facilitating decentralized cooperative optimization. LMMCoDrive represents a significant milestone towards practical, efficient, and safe AMoD systems, which also offers insights for future urban transportation solutions. In terms of our future work, we plan to reduce latency in multimodal data processing by leveraging the MoE technique and ensure real-time performance in large urban environments by ADMM.

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